

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	534648
Award Program	Seeding Solutions, Food Waster and Loss
Project Title	A novel desiccant system enables energy-efficient drying to reduce postharvest loss of agricultural commodities and foods
Grant Start Date	06/01/2017
Grant End Date	05/31/2022
Total Grant Budget	\$1,580,814.00
Total FFAR Budget	\$790,407.00
Matching Funder(s)	UC Davis Innovation Institute for Food and Health (IIFH)
Final Report Date	08/30/2022
Project Director/Principal Investigator Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Research was conducted at the Biological and Agricultural Engineering Department, at the University of California, Davis (UC Davis), CA, 95616.

- 1.2. How many new jobs were created because of this grant funding?

Table 1 describes the jobs created because of FFAR grant funding for each award year.

Table 1. Jobs created because of FFAR grant funding

Year	Jobs created
2018-19	1 Postdoctoral scholar: MD Abdul Momin. 1 Senior design developing engineer (part-time as part of the Bio and Ag Engineering Shop): Jedediah Roach. 2 Undergraduate students: Lucine Grigoryan and Carlos Orozco.
2019-20	1 M.S. student: Alice Dien.
2020-21	1 M.S. student: Laudia Anokye-Bempah. 1 Ph.D. student: Alice Dien.
2021-22	1 Ph.D. student: Tianyu Ying.

- 1.3. How many jobs were maintained because of this grant funding?

The postdoctoral scholar worked on the project from 2017 until 2020. Both undergraduate students graduated and left the project in 2021. All other jobs were maintained until the end of the grant period.



- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No changes.

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No changes.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

In 2019, the FAO reported that 14% of the food produced globally is spoiled before reaching the consumer. Unlike food waste, food loss occurs at the post-harvest and processing stages due to inadequate handling and storage. Losses of fresh fruits and vegetables receive the most attention for targeted postharvest solutions, but commodities such as cereals, oilseeds, and pulses account for more than 60% of the food calories lost globally and most losses occur because of ineffective drying. This project investigated the use of desiccants as an alternative drying and enhanced-storage method.

We built a desiccant-drying prototype to study the performance of Drying Beads™ (a zeolite-based desiccant) when used to dry rice, almonds, and walnuts. The drying process using desiccants was equally effective and had greater potential for energy and economic savings than heated-air drying. More specifically, desiccant-drying would allow for optimization in three domains that are currently inaccessible when using heated-air drying: 1) energy efficiency, 2) fuel choice, and 3) time-of-use flexibility.

From a packaging and storage perspective, desiccants proved to maintain the optimum storage conditions within hermetic packages during green coffee storage and transportation. Considering that coffee is the second most traded agricultural commodity after crude oil, the impact of these findings could have significant economic implications.

Desiccants have shown to be a promising alternative to prevent quantitative and economic losses throughout the U.S. and the world's food system. Scaling up this



technology could enable producers, processors, distributors, and consumers to achieve superior product quality with greater energy efficiency and reduce their processing costs.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Our project has proven that desiccant-drying could be a promising alternative to current agricultural drying and packaging methods. We tested the use of convective desiccant-drying as an alternative method to conventional heated-air drying when drying corn, rice, and tree nuts. In all experiments, we observed that although there was no significant improvement in product quality, the drying process was equally effective and had greater potential for energy and economic savings than heated-air drying. More specifically, desiccant-drying would allow for optimization in three domains that are currently inaccessible when using heated-air drying. These domains are 1) energy efficiency, 2) fuel choice, and 3) time-of-use flexibility.

1) Drying energy efficiency

The desiccants we used during our study (Drying Beads™) released heat during the drying process. Through this mechanism, not only were we able to get air with a relative humidity close to 0% but the drying process was enhanced by the heat released by the desiccants. On the other hand, the regeneration process of the desiccants produced a stream of hot air from which energy can be recaptured. The regeneration temperature is significantly higher than the temperatures seen during a traditional heated-air treatment. Thus, the potential for energy recirculation and system efficiency are high. Overall, this would mean that the energy cost of drying to farmers and processors would be reduced.

2) Fuel choice

By using desiccant systems, we can replace using non-renewable energy sources (i.e., natural gas and propane) with electricity. This flexibility in the fuel choice is important from an economic and from an environmental standpoint.

For instance, with the natural gas price increase experienced in 2022, farmers could choose to switch their main source of energy to electricity and reduce the energetic cost of their drying operation. Similarly, a desiccant system could be plugged into solar panels and operate off-grid.

From an environmental standpoint, using electricity is the easiest way of replacing fossil fuel energy with renewable energy sources. However, this would imply that if the system is connected to the grid, we would need to ensure that

the regional electric grid where it is plugged is a low-carbon electric mix.

3) Time-of use

The time-of-use refers to the moment in which the highest amount of energy needs to be used. With current heated-air dryers the drying air temperature needs to be maintained constant throughout the entire drying process. Sometimes, this means turning on the dryer and leaving it on 24h a day for several months.

The drying process with desiccants only requires the energy supplied to the fan, as most of the energy utilization (>85 %) is used for the regeneration process. Since the regeneration process and the drying can be performed separately, the drying operator can decide when to regenerate the desiccants, and therefore, when to use the energy. Thus, if the regeneration process is performed in times of a day when the electricity grid has an abundance of renewable energy the price and environmental impact of the drying process could be offset.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways: a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used; b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential outcomes: Our project demonstrated the desiccant’s ability to dehydrate rice, corn, almonds, and walnuts. Grains and tree nuts are some of the most important commodities in the U.S. agriculture. Convective desiccant-drying could reduce the cost related to processing these crops and increase farmers’ revenue. These results were particularly satisfactory with rice drying, where we found that drying time was decreased when using desiccant-drying.

In addition, our studies showed that storage of green coffee in hermetic packages using desiccants helped maintained optimum storage conditions, especially in extreme environmental conditions. Given that green coffee with a moisture content above 12% is prone to degradation, the use of desiccants can play an important role in eliminating moisture damage and preserving the quality of green coffee, therefore preventing quantitative and economic losses. Other commodities, including cocoa, maize, rice, beans, and walnuts are also prone to moisture intake during transportation. Thus, incorporating desiccants into hermetic packaging could be beneficial to the food import and export industry.



Intermediate outcomes: The results from this project help inform energy-related policy. For instance, drying of walnuts and rice in the U.S. mostly happens in California. With the California's commitment to switching to a "zero-carbon" economy, technologies that enable fuel switching and/or time-of-use flexibility such as desiccant-drying will be needed. The findings from our studies are the first step towards designing and optimizing such systems.

We have discussed our findings with the specialty coffee association and have also presented our results in their yearly meetings (as seen below), and companies producing hermetic packages. When feasible and required, the industry will use desiccants during storage and transportation to better preserve the moisture of green coffee.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The project had 4 major milestones as described below. Table 2 shows the specific objectives and tasks related to each milestone and the level of completion.

Milestone 1. A prototype drying system will be designed, modeled, constructed, and instrumented to determine the impacts of a range of system configurations (e.g., air flows, reactivation methods, type and variety of produce, RH, etc.) on total drying efficiency. To calculate energy efficiency, the specific heat consumption will be calculated by dividing the energy supplied to the drier by the mass of water evaporated from the produce. Compare the energy use and performance of different systems (e.g., crossflow, mixed-flow, concurrent-flow, and counter-flow systems) independently and with the Drying Bead technology integrated into these same systems.

Milestone 2. Bead effectiveness in the drying system on the macroscopic level (thermodynamic properties) will be assessed with respect to the key processes identified in Figure 4. Time required for Drying Beads to reach their water-holding capacity (1) will be determined at different input air RH, temperatures, and air flow rates. In this process, the exothermic reaction (heat generation) by the Drying Beads upon water absorption (2) will be quantified. Power inputs, temperature and duration required to reactivate (dehydrate) Drying Beads (3) will be determined for various types of heaters and flow-through configurations. The continuous flow design provides significant opportunities to recapture the heat used in their reactivation through heat exchangers to preheat incoming Drying Beads while cooling Drying Beads exiting the reactivation heater (4). Similarly, the air exiting the reactivation heater to exhaust released moisture can be used to preheat air entering the heater (5). We will quantify the magnitudes of these heat recovery processes and their impacts on overall energy efficiency. We will capture technical data on drying effectiveness, energy consumption and feasibility for scaling up.



Milestone 3. With the quantitative data on Drying Beads and system behavior, models will be constructed to enable testing of scaled-up configurations. The most efficient systems for different applications will be designed and modeled.

Milestone 4. We will conduct a techno-economic assessment of the optimized Bead-based system relative to conventional technologies. We will directly compare the leveled cost (integrated capital and operational expenditures per unit of product) of the Bead technology based on our modeled system relative to existing data on conventional systems by crop type. A linked evaluation will specifically explore the potential process energy savings (per unit product by crop type) between the proposed technology and conventional systems from a life-cycle perspective. Finally, using existing data on food losses and waste linked to insufficient or ineffective drying, we will estimate the potential reduction of food waste enabled by the Bead technology at the global scale if fully deployed to underserved markets. To the extent that data are available, we will then derive the linked land, water, and energy resources conserved through these avoided losses in food products.

Table 2. Project milestones, objectives, tasks, and level of completion.

Milestone	Objective	Task	Status	Level of completion
Milestone 1 Desiccant Drying Performance	Objective 1.1. To evaluate the performance of the FlexiDry drying system in terms of drying time, drying rate and energy efficiency.	T1. System acquisition.	Completed March 2018	100%
		T2. Drying of corn using heated and unheated air, comparison with conventional heated-air drying.	Completed June 2019	100%
	Objective 1.2. To build a system that allows to test desiccant-drying at different ambient conditions and with different commodities	T3. Design of the Vertical Dryer (VD).	Completed September 2018	100%
		T4. Build the dryer system.	Completed September 2019	100%
	Objective 1.3. To compare the energy consumption of the Drying Beads with other drying systems for a specific crop.	T5. Estimate the energy consumption of experiments performed using the chimney dryer.	Completed March 2022	100%
		T6. Perform a literature review to find studies that report energy consumption for drying of commodities used in the experiments.	Completed June 2019	100%
	Objective 1.4. To compare the drying performance (drying rate, time, bead ratio, etc.) of desiccant drying compared to conventional drying for different crops.	T7. Perform experiments using different crops and different drying conditions (RH, bead ratio, etc.). Use convective heated air drying as the reference	Completed October 2021	100%
		T8. Collect data on quality parameters after drying (specific for each crop).	Completed October 2021	100%
	Objective 1.5. To determine the feasibility of using desiccants	T9. Understand the suitable conditions for coffee storage.	Completed October 2021	100%

	to achieve optimal storage conditions for produce using coffee as a model crop.	T10. Use Drying Beads to provide appropriate storage conditions for coffee storage.	Completed October 2021	100%
	Objective 2.1. To find the time required for Drying Beads to reach their water-holding capacity at different RH, airflows, and temperatures.	T11. Perform experiments using Drying Beads and FlexiDry desiccant beads.	Completed April 2021	100%
	Objective 2.2. To quantify the heat released by drying beads upon water adsorption.	T12. Use the data from experiments to quantify the amount of heat released by drying beads upon water adsorption.	Completed March 2022	100%
		T13. Build a system that allows accurate regeneration modeling and testing.	Expected September 2022	80%
Milestone 2 Drying Beads and Desiccant Drying	Objective 2.3. To determine the power input, temperature, and duration required to regenerate the beads.	T14. Perform experiment using different temperatures and times.	Expected December 2022	0%
		T15. Calculate the amount of energy used per regeneration and per kg of beads regenerated.	Expected December 2022	0%
	Objective 2.4. To determine the potential of energy recovery from the regeneration process	T16. Use the data from experiments to calculate the amount of energy the air would hold if it was exhausted during regeneration.	Expected December 2022	0%
	Objective 2.5. To model and validate the regeneration process.	T17. Use a physical model to predict the behavior of Drying Beads when regenerated in a convective oven at 250C.	Completed June 2020	100%
		T18. Conduct experiments to validate the model.	Expected September 2022	30%

Milestone 3 Scaled-up configuration	Objective 3.1. To model the regeneration process of multiple beads.	T19. Revise the model based on empirical evidence.	Expected December 2022	0%
		T20. Use a physical model to predict the behavior of Drying Beads when regenerated in a convective oven at 250C.	Expected September 2022	60%
		T21. Conduct experiments to validate the model.	Expected September 2022	0%
	Objective 3.2. To simulate different regeneration configurations and bead rations and calculate their impact on energy consumption and design.	T22. Revise the model based on empirical evidence.	Expected December 2022	0%
		T23. Use mathematical models to predict how different regeneration configurations would affect the regeneration process.	Expected June 2023	0%
Milestone 4 Techno-economic, LCA and Community Impact	Objective 4.1. To perform a techno-economic assessment of a bead-based technology relative to conventional technologies.	T24. To compare the cost (integrated capital and operational expenditures per unit of product) of bead technology relative to existing drying technologies.	Expected September 2023	50%
		T25. Perform an energy balance considering energy recirculation options and re-estimate the energy consumption and cost of an optimized system using the models developed in objective 3.2.	Expected September 2023	50%
		T26. Re-calculate the cost using alternative energy sources (e.g., solar)	Expected September 2023	0%
	Objective 4.2. To assess the environmental impact of desiccant-drying compared to	T27. To perform a comparative life cycle assessment (LCA) for Drying Beads and heated air drying when used in rice.	Expected December 2022	80%



conventional drying
technologies.

Objective 4.3. To assess the
food safety and security
improvement of desiccant-
drying.

T28. To estimate the potential
reduction of food waste enabled by the
bead technology at the global scale if
fully deployed to underserved markets.

Expected September
2023

0%



- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (**up to 500-word limit**)

No major changes.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (**up to 5,000-word limit**)

Table 3 summarizes the major objectives (from table 2) completed under each milestone, a description of each objective, the key findings, and the key achievements. Activities leading to major findings and achievements are further detailed later in this section.



Milestone	Major objectives	Description	Significant results	Key outcomes
Milestone 1 Desiccant Drying Performance	Objective 1.1. To evaluate the performance of the FlexiDry® drying system in terms of drying time, drying rate and energy efficiency.	We acquired an existing industry-scale desiccant-drying system and we tested its performance when used to dry corn.	Although we expected to get RH values close to 0%, the RH values were never below 10%. It seemed like the regeneration of the desiccants was never fully achieved and we concluded that the design of the system was not optimized.	We used what we learned and the unanswered questions we had when testing the FlexiDry® to brainstorm which questions an ideal lab-scale system should answer and how to design it.
	Objective 1.2. To build a system that allows to test desiccant-drying at different ambient conditions and with different commodities	We designed and built a lab-scale convective dryer that allows us to test different convective drying regimens including desiccant-drying, heated-air drying and ambient-air drying. The system records the necessary data for subsequent calculations and conclusions.	Most of the project time went into designing and building this system. Complicated engineering problems had to be solved for this system to fully capture the drying process and enable the necessary measurements. Multiple iterations of the system were built before we obtained a definitive system.	With this system we can study the drying performance of any convective regimen while simulating the ambient conditions encountered anywhere in the World. For instance, we used the system to test drying rice in California vs Thailand environmental drying conditions.
	Objective 1.3. To compare the energy	We dried almonds, walnuts, and rice with Drying Beads™ and	We found that as it stands, desiccant-drying uses more	Despite consuming more energy, desiccant-drying has more design flexibility than

consumption of the Drying Beads with other drying systems for a specific crop.	heated air drying and compared the energy consumption of one drying method to the other.	energy than heated-air drying.	heated air drying. Thus, it is equally important to study the potential of energy recirculation, crop quality, fuel switching, and time of use of energy.
Objective 1.4. To compare the drying performance (drying rate, time, bead ratio, etc.) of desiccant drying compared to conventional drying for different crops.	We dried almonds, walnuts, and rice with Drying Beads™ and heated air drying, and we compared the drying rate, drying time and bead ratio of one drying method to the other.	We found that the drying time of almonds and walnuts was longer with desiccants compared to heated air drying. In contrast, rice was dried faster when using desiccants, especially at high ambient RH.	The results suggest that desiccants might be more effective and/or efficient when used to dry grains (e.g., corn, rice) rather than tree nuts.
Objective 1.5. To determine the feasibility of using desiccants to achieve optimal storage conditions for produce using coffee as a model crop.	We placed green coffee in jute bags and hermetic bags with and without desiccants (CaCl ₂ and Drying Beads™). We stored the coffee in environmental chambers simulating typical storage and transportation conditions. We wanted to see if the presence of desiccants helped keep optimum moisture conditions.	Desiccants were helpful with storage in extreme environmental conditions (e.g., maritime and railway transportation) where they prevented the coffee from absorbing excess moisture. However, the packaging needs to be hermetic otherwise desiccants equilibrate too fast with the environment and no difference in MC between packaging is observed.	The results show that hermetic packaging with desiccants is a viable option for improving transportation and storage of coffee. We believe this technique could even be implemented as a way to reduce the MC of coffee during transportation and accelerate the drying process of coffee.



Milestone 2 Defining Properties of Drying Beads	Objective 2.1. To find the time required for Drying Beads to reach their water-holding capacity at different RH, airflows, and temperatures.	We investigated how different environmental conditions might affect the desiccant behavior and their ability to dry crops.	We found that desiccants drop the RH of air close to 0% and then, air slowly increases back up to ambient RH. The higher the RH in the ambient air, the fastest this happens.	We can achieve longer drying periods when ambient conditions are at a low relative humidity.
	Objective 2.2. To quantify the heat released by drying beads upon water adsorption.	We measured the temperature of the air upon water adsorption by the drying beads to quantify the amount of heat released by the desiccant.	We found that the air temperature increases at higher ambient RH values, which would suggest that Drying Beads have a stronger exothermic reaction in these conditions.	Although lower RHs increase the drying time of the beads, it seems that when the exothermic reaction is considered beads display a better drying performance than heated-air drying.
	Objective 2.3. To determine the power input, temperature, and duration required to regenerate the beads.	We built a system using an existing coffee roaster that will allow us to quantify the amount of heat required to regenerate the Drying Beads at different temperatures.	Since we are still building the regeneration equipment, we do not have results on the regeneration portion yet.	The data obtained with this system will allow us to calculate the energy consumption of the experiments we conducted and validate the model from objective 3.1.
Milestone 3 Scaled-up configuration	Objective 3.1. To model the regeneration process of multiple beads.	We are building a model that will allow us to predict how different regeneration parameters affect the regeneration time and energy consumption.	The model will be validated with the results from Objective 2.3. and will be used to make predictions for varying regeneration configurations.	With this model, we can predict the behavior of a full-scale desiccant dryer and perform the necessary LCA and techno-economic analyses.



	Activity 3.4 Determining the potential of energy recovery from the regeneration process.	We will measure the temperature of the exhaust air during the regeneration process and use an energy balance on to calculate the potential energy recovery.	With the energy recovery potential, we can calculate the overall system efficiency.	We will use this information to predict how energy recovery influences the overall performance of the system and what are the potential economic savings that can derive from it.
Milestone 4 LCA and Community Impact	Objective 4.1. To perform a techno-economic assessment of a bead-based technology relative to conventional technologies and compare it to conventional drying.	We will perform a techno-economic assessment to evaluate the economic feasibility of desiccant-drying and determine potential cost savings compared to heated-air drying.	We are currently building the techno-economic analysis and creating an inventory with all the process input-outputs.	The techno-economic model will show the potential economic savings of desiccant-drying compared to heated-air drying. The sensitivity analysis will highlight potential areas of improvement that will yield the biggest economic savings.
	Objective 4.2. To assess the environmental impact of desiccant-drying compared to conventional drying technologies.	We will assess the environmental impact from desiccant-drying compared to heated-air drying using a comparative life cycle assessment (LCA).	We are currently building the LCA model with the input/outputs of the technology and creating an inventory. However, we are still missing some data from previous objectives to fully finalize this phase.	The environmental LCA will show a comparison of the environmental impacts of desiccant-based drying and heated-air drying, especially in terms of global warming potential.

Objective 1.1: Testing the FlexiDry®

The first activity our team performed when the FFAR project started was to partner with AgInnovation USA (<https://www.aginnovationusa.com>) and Rhino Research (<http://www.rhino-research.com>) who provided us with an existing industrial-scale seed desiccant dryer (FlexiDry®) and the desiccant s (Drying Beads™) to test the dryer.

We evaluated the performance of the FlexiDry in drying 130 kg of whole yellow corn kernels (*Zea mays*) for large-scale human and animal consumption. FlexiDry was tested in two treatments, using heated (HA) and unheated (UA) air, and compared to conventional heated-air (CHA) drying. The Drying Beads were regenerated at 250 °C using a furnace within the FlexiDry system.

As shown in figure 1. All treatments properly dried the corn below the target value of 15% dry basis moisture content (M_D). The CHA and HA treatments yielded an average moisture extraction rate (MER) of 1.42 kg/h, compared to 0.58 kg/h for the UA treatment.

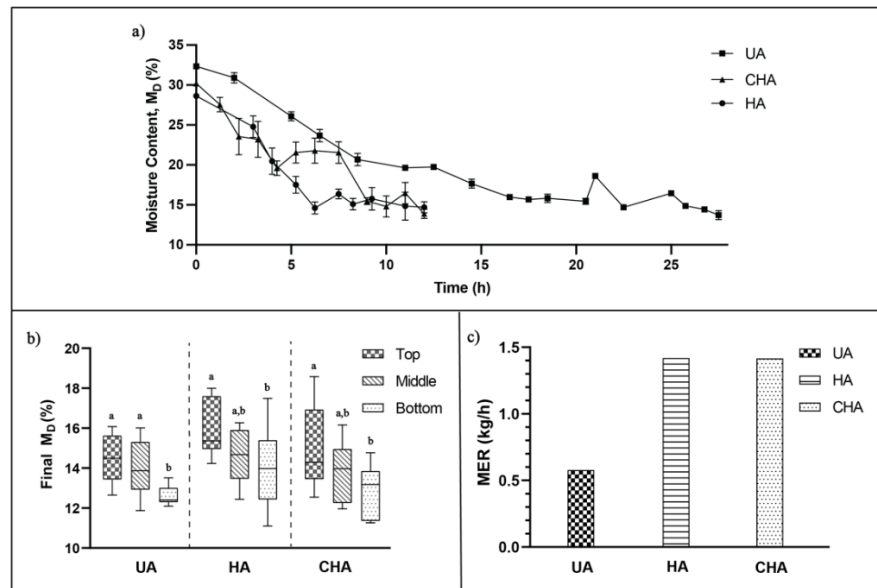


Figure 1. (a) M_D over time with standard errors of mean (SEM). (b) Boxplots showing final M_D for each treatment at the top, middle, and bottom layers within the drying barrel. (c) Moisture Extraction Rate (MER) values for each treatment.

In terms of energy consumption (figure 2), the specific moisture extraction rate (SMER) was 0.2, 0.44, and 1.11 kg/kWh for the UA, HA, and CHA treatments, respectively. The coefficient of performance (COP), i.e., the ratio between the

theoretical heat delivered and the total energy consumed, was 0.14, 0.30, and 0.75 for the UA, HA, and CHA treatments, respectively. The FlexiDry treatments (HA and UA) consumed more energy than the CHA treatment, with more than 85% of the energy consumption used to regenerate the Drying Beads.

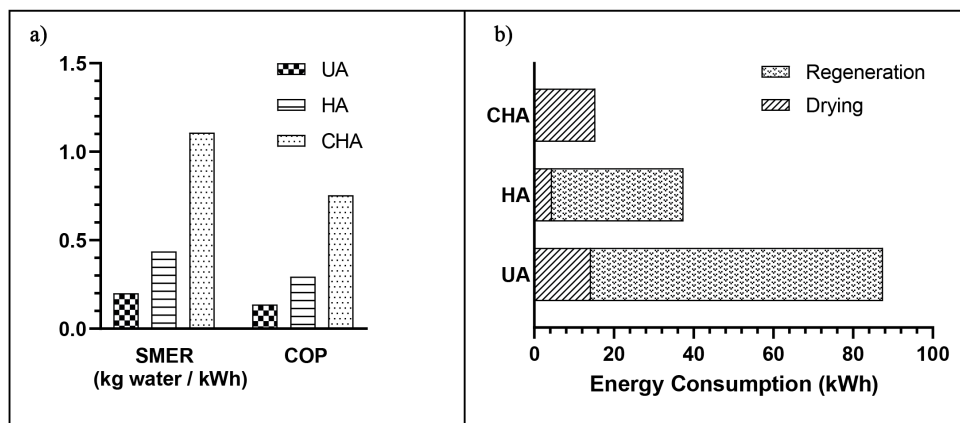


Figure 2. (a) Specific moisture extraction rate (SMER) and coefficient of performance (COP) for each treatment and (b) total energy consumption for each treatment split between regeneration and drying.

The system yielded a global warming potential (GWP) of 0.931 and 0.425 kg CO₂-eq/kg of water removed for the UA and HA treatments, respectively (figure 3). These values are higher than the GWP of the U.S. con industry drying standard, which is reported as 0.383 kg CO₂-eq/kg of water. Thus, if the regeneration process can be optimized, there is potential for reductions in energy intensity and GWP with large-scale, desiccant-based drying of agricultural commodities.

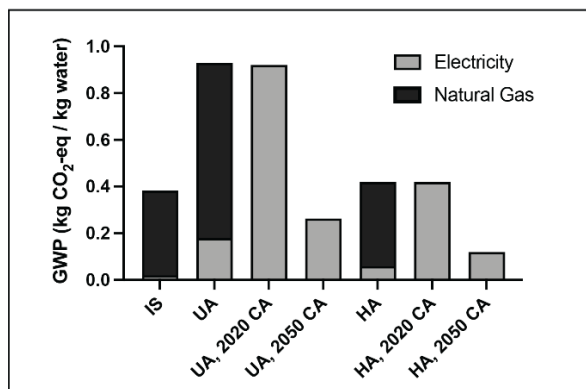


Figure 3. Global warming potential (GWP) for each treatment: IS: Industry Standard. UA: UA treatment. UA, 2020 CA: UA treatment after fuel-switching from natural gas to electricity with the 2020 CA mix. UA, 2050 CA: UA treatment after fuel switching from natural gas to electricity with the 2050 predicted CA mix. HA: HA treatment. HA, 2020 CA: HA treatment after fuel-



switching from natural gas to electricity with the 2020 CA mix. HA, 2050 CA: HA treatment after fuel switching from natural gas to electricity with the 2050 predicted CA mix.

The FlexiDry® system effectively dried 130 kg of corn from an average initial M_D of 30.4 (± 1.6) % to an M_D of 14.1 (± 1.7) %, which was below the target value of 15 % (d.b.). However, in both UA and HA drying treatments, the system was more energy-intensive than the lab-scale CHA system and the IS, especially when operating under the UA treatment. Similarly, the GWP of both desiccant treatments was higher than the IS. Reducing GHG emissions from desiccant-drying systems would require improving the FlexiDry design, the energy efficiency of the regeneration process, fuel switching from natural gas to electricity, and performing the regeneration in an electric grid mix that has a high percentage of renewable energy.

It was expected that with the Drying Beads, the drying air RH would be equal or close to 0 %, at least during the beginning of the drying process. However, the RH was never below 10 %. Since the Drying Beads verification experiments did not show a significant difference between the FlexiDry and newly purchased Drying Beads, the observed discrepancies could be due to the FlexiDry design. However, numerous questions regarding the optimization of large-scale drying and regeneration of desiccants in agriculture remain unanswered and need to be investigated. For this reason, we built a laboratory-scale desiccant-based dryer capable of monitoring and further exploring the drying process using desiccants. Particularly, the system can control the inlet air conditions (T and RH) to determine how desiccant performance could vary across initial conditions, locations, and seasons.

Objective 1.2: Building a laboratory-scale system

We designed and built a research-scale deep-bed forced convection dryer (Vertical Dryer (VD) system, see Figure 4) to study the performance of three convective drying regimes: ambient air drying, heated air drying, and ambient air forced through a desiccant material. The VD was designed to perform studies that would help optimize current large-scale commercial agricultural dryers and evaluate desiccant drying techniques.

The VD's inlet air can be controlled to achieve desired environmental conditions (temperature (T) and relative humidity (RH)), and airflow. The system is divided in three sections: 1) An air delivery system that provides the conditioned air (airflow, temperature, relative humidity); 2) A modular section to perform and monitor the drying process; 3) A software and data collection system that records real-time data on the drying process.

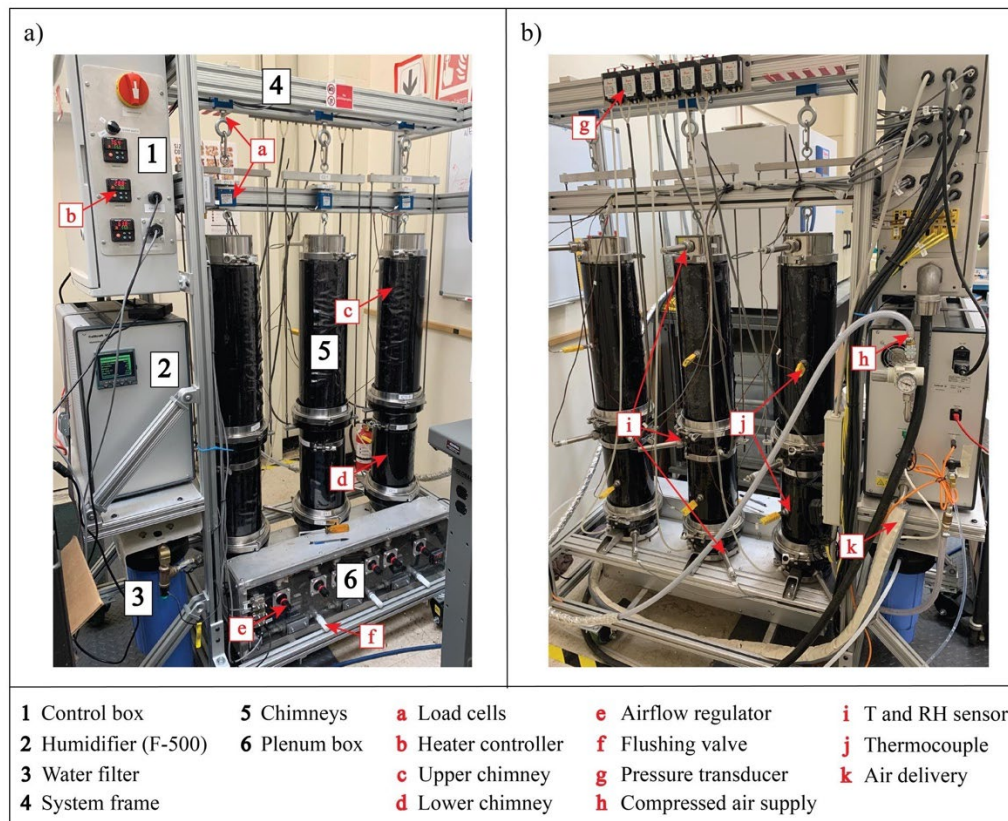


Figure 4. Laboratory-scale dryer (Vertical Dryer) and its main components.

Objective 1.4: Drying performance

Rice

We assessed the drying performance and quality of Calrose rice when dried with Drying Beads®. The drying process utilized a rice paddy to DB ratio of 1:2 and was compared to conventional heated-air drying (HA) at 50°C. Experiments were performed at two typical paddy drying environments (California and Thailand) with an airflow of 115 L/min, ambient air temperature and relative humidity (RH) of 22°C and 45% RH (DB1), and 27°C and 80% RH (DB2). All of the drying experiments involved dehydrating paddy to a wet basis moisture content (MC_{wb}) of 14 % or less as shown in figure 5.

The moisture removal rate (MRR) for DB1 and DB2 were 9.3% lower and 84.3% higher in comparison to HA, respectively. Head rice yield (HRY) for DB1 and DB2 were 10% higher and 27.5% lower in comparison to HA, respectively. DB1, DB2, and HA produced similar total milling yields (TMY) of over 71%, and the rice kernel color did not differ significantly. This study demonstrates that utilizing DB for rice paddy drying is promising in a highly humid environment (DB2), but an improved system may be required to increase its HRY.

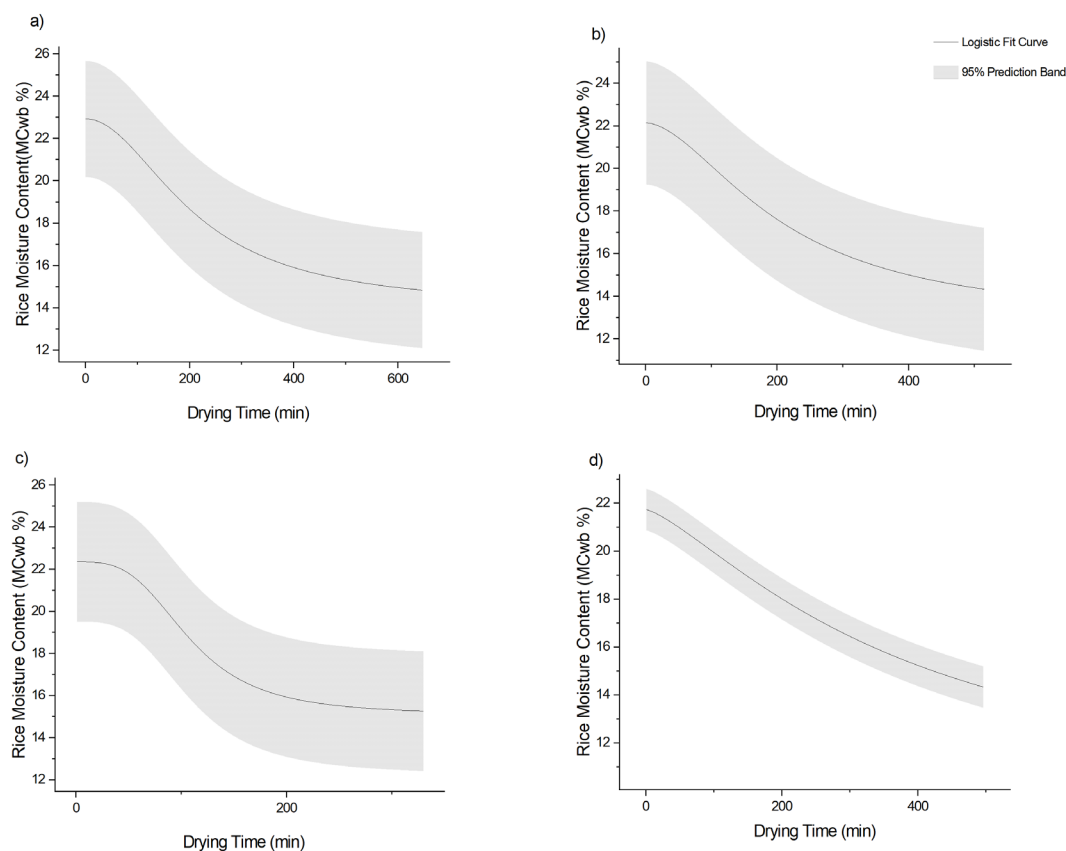


Figure 5. Best fitting drying curves with 95% confidence interval for a)DB1, b)HA1, c)DB2, d)HA2.

Objective 1.5: Coffee storage

We investigated the efficacy of commercially available desiccants for preserving the moisture content of green coffee between its optimal range (10-12%) when stored in either hermetic packages and/or jute sacks and assessed the corresponding impact on sensory quality. Green coffee was packed in three packaging materials (GrainPro SuperGrain bag, GrainPro TranSafeliner, and/or jute sacks), with and without desiccants (Drying Beads™, and CaCl₂) and stored for 42 days in environmental chambers under conditions of fluctuating T and RH that mimic typical storage and transportation conditions. The moisture content, water activity, and color of all samples were measured approximately weekly over 42 days.

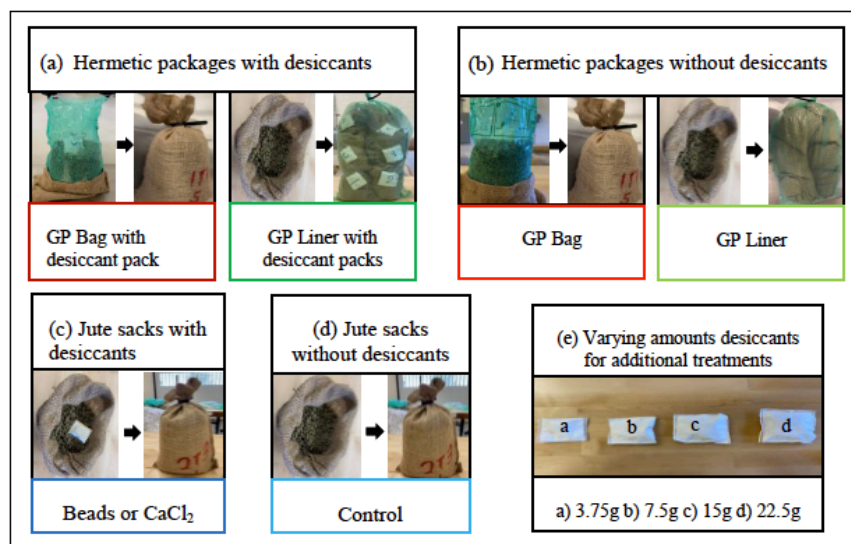


Figure 6. Schematic representation of treatments and preparation procedures.

Overall, the results indicated that hermetic packages, even without desiccants, provided a stable moisture content for green coffee compared to jute sacks only, but under extreme conditions such as during green coffee transportation, desiccants should be added to prevent the moisture content to increase above 12% (see figure 7). In comparison to the control (no desiccant, and only jute sack), we observed a statistically significant impact for all tested desiccants and hermetic packages for maintaining the proper moisture content, water activity, and color. No significant difference was observed for the different desiccant masses tested when they were placed inside the hermetic packaging, but the desiccants were ineffective without the hermetic packaging. The triangle test and descriptive sensory evaluation yielded no significant differences between the use of hermetic packages with desiccants, and without desiccants.

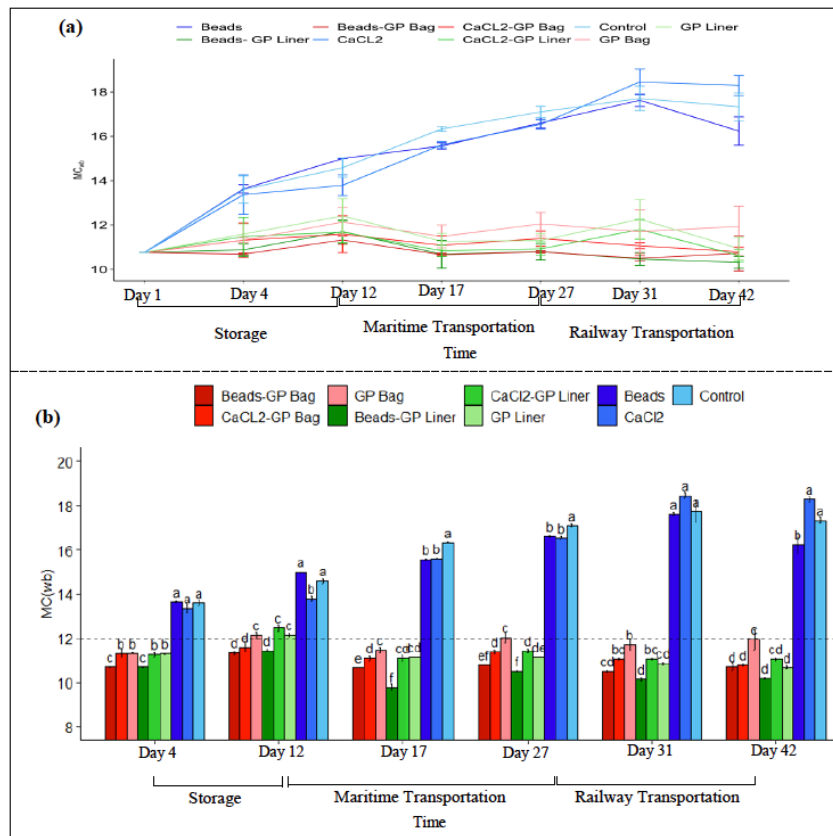


Figure 7. a) Line graph showing the variation in moisture content for each treatment during green coffee storage and transportation. b) Effect of treatments on moisture content during transportation.

Objective 2.5: Model Regeneration Process

We modeled the regeneration process of Drying Beads™ to understand how beads behave when placed under a stream of hot air. The model considered a single-bead and analyzed the temperature and water distribution in the beads over time.

The partial differential equations (PDEs) are:

PDE Mass Transfer	PDE Heat Transfer
$\frac{\partial X}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 D(T) \frac{\partial X}{\partial r}$	$\frac{\partial [C_v(T, X) T]}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 k(X) \frac{\partial T}{\partial r}$
Where: r = Bead radius [m] D (T) = Diffusion coefficient [m/s] X = Massic fraction of water [kg water/kg bead] t = time [sec]	Where: r = Bead radius [m] k (X) = System thermal conductivity [W/m.°C] C _v (T,X) = System volumetric heat capacity [J/m ³ .°C] T = Bead temperature [°C] t = time [sec]

The boundary conditions (BCs) and Initial Conditions (IC) are:



BCs for mass transfer		
$r=a$	$-D(T)\frac{\partial X}{\partial r} = k_m(X - \frac{C_\infty}{\rho_b})$	Where: k_m = Mass transfer coefficient [m/s] C_a = Concentration of water at $r=a$ [kg water/ m^3] C_∞ = Concentration of water in the drying air [kg water/ m^3]
$r=0$	$\nabla C = 0 \longrightarrow \frac{\partial X}{\partial r} = 0$	
BCs for heat transfer		
$r=a$	$-k(X)\frac{\partial T}{\partial r} = h(T_a - T_\infty)$	Where: h = Convective heat transfer coefficient [W/ m^2s] T_a = Temperature at $r=a$ [$^{\circ}C$] T_∞ = Temperature of drying air [$^{\circ}C$]
$r=0$	$\nabla T = 0 \longrightarrow \frac{\partial T}{\partial r} = 0$	
Initial Condition (IC)		
$t=0$	$X = 0.25 \frac{kg \text{ water}}{kg \text{ bead}}$ $T = 25 \text{ }^{\circ}C$	

Figure 8 represents the mean temperature (blue curve) and water concentration (red curve) over the bead at each time step. We can broadly characterize the temperature curve in three main parts. First, between 0 and 20 seconds, the temperature rises steadily from 25 °C to 100 °C. At 100 °C, water begins to evaporate. As a result, the temperature stops rising and attains a plateau (see figures 9a and 9b) where most of the heat supplied to the bead is used as vaporization energy. Once the water has completely evaporated ($t=40s$), the temperature in the bead quickly raises again until the entire bead is at 250 °C (air temperature).

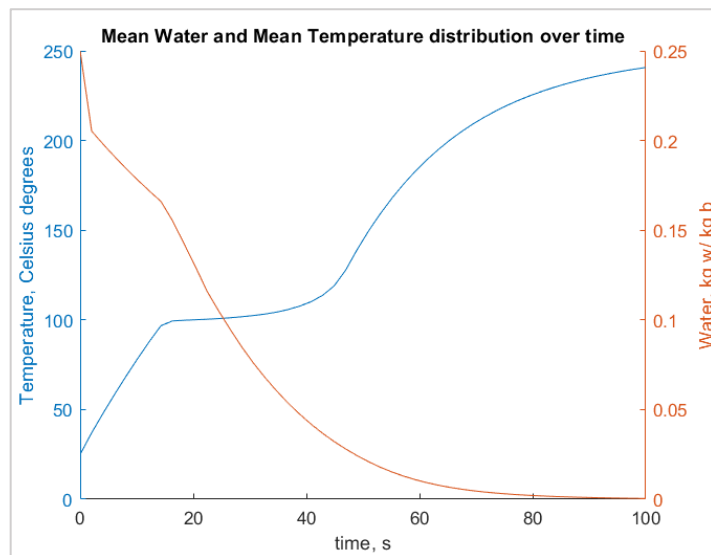


Figure 8. Mean water and temperature distribution.

The water fraction starts at a dried basis moisture content (M_D) of 25% and rapidly falls to 20% (M_D). Then, it decreases steadily to reach 17% (M_D). At this point, the

bead has reached 100 °C, water becomes vapor and is quickly being expelled. From figures 2c and 2d, we observe a faster release of water from the outer layers than from the inner ones. We also observe a discontinuity between the layers that are above and below 100 °C. The reason behind the discontinuity is that as soon as the temperature is above 100 °C, liquid water cannot subsist, the diffusion coefficient becomes very large, and the vapor water is directly expelled from the bead. Overall, the model is consistent with the trends we observed in empirical experiments.

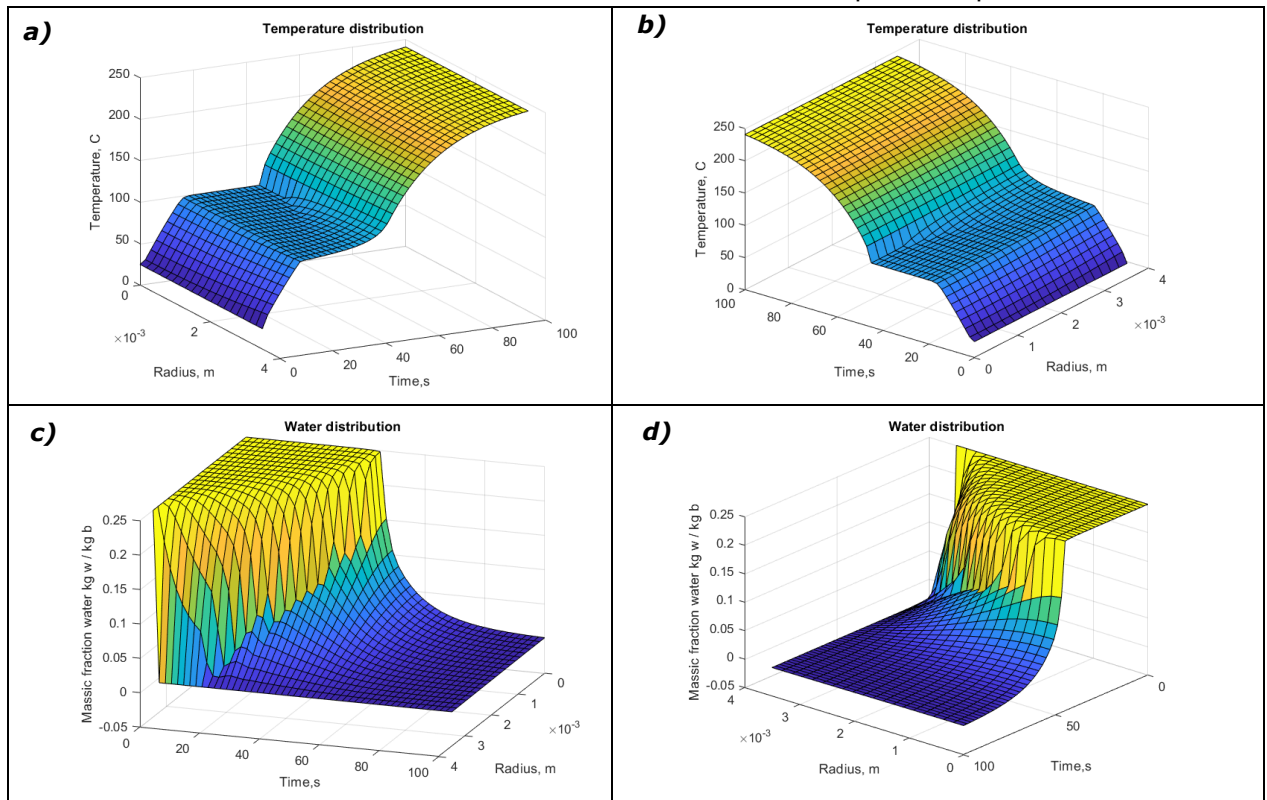


Figure 9. 3-D plot of water and temperature distribution.

For this model, we considered a single bead with uniform convection on its surface. However, when performing the regeneration process in real life, beads are packed. As a result, the convection around the bead will not be as uniform, and we might see higher differences in temperature distribution on the radial direction of the bed. Besides, the heat transfer and mass transfer inside the bead will be slower, so the increase in temperature and moisture reduction within the bead will take longer. The model was built considering that there is only liquid water inside the bead. Further model iterations should account for the fraction of vapor water in the bead and better define the diffusion coefficient of vapor. Furthermore, the heating time is highly dependent on the convective coefficient (h), and the drying time is highly dependent on the mass transfer coefficient (k_m). Therefore, D , h , and k_m should be accurately defined or checked with experimental data.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include



approaches taken to ensure robust and unbiased results.

At the beginning of the project our team wanted to understand the challenges existing in the desiccant-drying industry of agricultural commodities. To do so, we partnered with AgInnovation USA, and Rhino Research and they provided us with an existing industrial-scale seed desiccant dryer. We tested the system and looked at its performance objectively to assess what was the “state of the art” on this type of system and how to best allocate our research efforts so that we could best contribute to the industry and the U.S. agriculture.

Once this phase was completed, we conducted our own experiments using the laboratory-scale dryer we built in the lab. The laboratory-scale was not built as an industrial-scale system design, but rather to measure key parameters needed to further assess the feasibility of developing convective desiccant-drying systems.

Prior to each experiment, we conducted a literature review and an experimental design to ensure that our research approach was scientifically rigorous and that valuable results could be extracted and statistically analyzed from each set of experiments. Two examples of experimental designs we used include a completely randomized design (CRD) and response surface methodology (RSM). For each set of experiments, statistical analyses were conducted using R software (ver. 4.0.3, R Core Team 2020, Vienna, Austria).

The main research team had weekly meetings (in-person or virtually) to check on the project progress and to discuss results, conclusions, and further steps.

When a set of experiments ended and the necessary analyses were completed, the results went through a “peer-reviewed” process in which all PIs involved in the project revised the work and provided feedback and recommendations.

2.8. List key stakeholders that could be served by results of this project.

Stakeholders that could benefit from the results of this project include tree nut and grain producers (including farmers), agriculture policy makers, energy policy makers, scientists working in the field of postharvest engineering and food processing, coffee farmers, coffee producers, agricultural equipment companies.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a

response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

2017-2018

The project was disseminated in multiple outreach and scientific presentations, including the following: 1) Annual meeting of the Walnut Grades and Standards Board (March 3rd, 2018); 2) Annual dry-chain workshop at UC Davis (April 20th, 2018); 3) 3rd annual innovator summit at UC Davis (April 23rd, 2018); 4) Postharvest technology short-course (June 18-22nd, 2018).

2018-2019

The project was disseminated in multiple outreach and scientific presentations including the following: 1) Annual meeting of the Walnut Grades and Standards board (March 07, 2019); 2) 4th annual innovator summit at UC Davis (May 22, 2019); 3) IFT19 annual meeting and food expo, New Orleans, LA (June 2-5, 2019); 4) Postharvest technology short course (June 17-21, 2019).



Figure 10. Ph.D. student Alice Dien presenting a poster on the project at the IFT19 meeting in New Orleans.

In August 2019, the project was disseminated during a meeting with all stakeholders to explain the results of the FlexiDry® study and discuss the next steps. This meeting occurred at the Western Center for Agricultural Equipment (WCAE) at UC Davis on August 8, 2019.



Figure 11. Left: Dr. Donis-González presenting at the FlexiDry stakeholder meeting at the WCAE, August 2019. Right: Dr. Momin showing the FlexiDry system to stakeholders.

2019-2020

2020-2021

The project was disseminated in multiple outreach and scientific presentations including the following: 1)

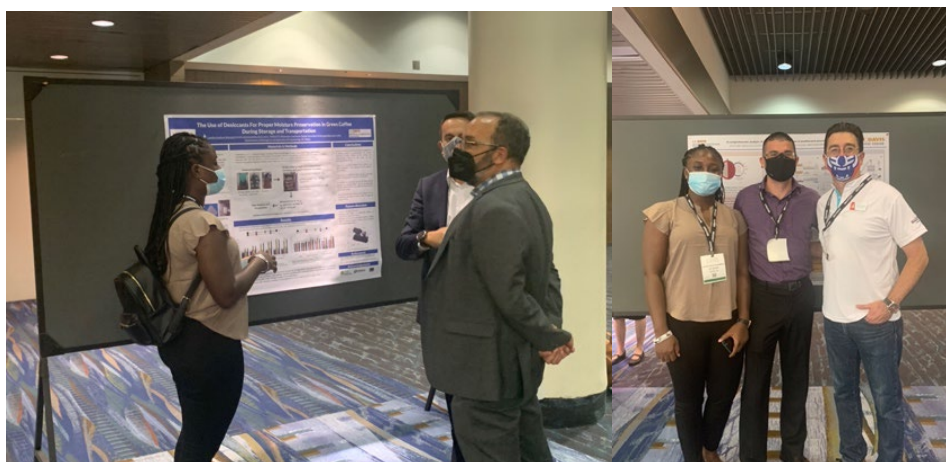


Figure 12. Left: M.S. student Laudia Anokye-Bempah explaining her research to Peter Giuliano, chief research officer of the Specialty Coffee Association. Right: Laudia and Dr. Donis-González.

2021-2022

On April 6th 2022, Ph.D. student Alice Dien won 1st place and Public Impact Award by pitching this project in a 3-min talk at the 2022 UC Davis Grad Slam competition which rewards students for communicating research to a general audience. Video and press-release can be found in the following link: <https://grad.ucdavis.edu/news/biological-systems-engineering-grad-student-alice-dien-wins-grad-slam>



Figure 13. Alice Dien with UC Davis Dean of Graduate Studies Jean-Pierre Delplanque receiving 1st prize at Grad Slam competition. Alice speaking at the UC system-wide Grad Slam competition.

On April 29th 2022, M.S. student Laudia Anokye-Bempah was awarded 1st Place in the Chicano and Latino Engineers and Scientists Society (CALESS) elevator pitch competition for pitching her research on Green Coffee Storage and Transportation.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

We had significant troubleshooting issues with the system we acquired at the beginning of the project (FlexiDry®). Firstly, it was challenging to find a place within UC Davis where we could run the system safely. Secondly, the system did not behave as specified by its manufacturers and we kept getting an alarm signal that would shut down the burner when trying to run the regeneration process. We had to troubleshoot the system to find the issue, and we ended up replacing the ignition system in both burners.

In 2019, postdoctoral fellow Md Abdul Momin who oversaw the project started a new position as a data analyst for the United States Department of Agriculture. Therefore, we had to restructure the work team. Alice Dien (a Ph.D. student in the project) took the leadership on continuing working on the project goals for the remaining of the grant period.

In 2020, the Covid-19 pandemic significantly affected our ability to conduct the research. While working closely with the university and the public-health mandates related to the pandemic we were unable to perform experiments from March 2019 to September 2019. Sensor manufacturing and other companies that provide us with pivotal supplies to complete our activities had significant delays in delivering and properly repairing these. Thus, we experienced significant delays in the manufacturing of the laboratory-scale desiccant system. On top of that, most of our



experiments depend on seasonal harvesting of fresh produce which further delayed some of our experiments. While working from home, each member of the group was assigned tasks that did not involve experimentation such as reviewing the literature, writing a program for the LabVIEW software, analyzing data that had already been collected and writing articles for publication. Weekly meetings were continued over Zoom in group or individually.

Troubleshooting the laboratory-scale dryer also took a significant amount of time as there were condensation issues happening within the system and leakages in some of the air delivery systems of the dryer. These leaks were fixed, and the condensation issues were resolved with dry runs. Similarly, we had to add insulation to some of the pipes carrying water and/or air into the system to ensure experiment conditions remained as constant and close to the set point as possible. One of the airflow meters malfunctioned during an experimental run. Unfortunately, the repair of the meter was delayed due to the Pandemic. We ended up purchasing new airflow meters from a different company.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

The undergraduate and graduate students involved in the project were trained in the following areas

Food Engineering

1) Drying bead regeneration and thermal properties; 2) Operation of conventional and desiccant-based dryers; 3) Mass and energy balances; 4) Mathematical modeling of biological processes; 5) Non-invasive food quality measurement techniques (color, bulk density); 6) Moisture content measurement; 7) Energy consumption calculations.

Hardware and Software

1) Designing and building engineering equipment; 2) LabVIEW programming for automatic data acquisition; 3) Use of "R" software for statistical computing and graphics; 4) Use of SolidWorks for 3-D drawing; 5) Configuration of sensors to acquire and monitor drying conditions.

Scientific Process and Research



1) Experimental design; 2) Statistical analyses; 3) Scientific rigor and unbiased results; 3) Scientific writing; 4) Manuscript publication process; 5) Poster preparation and presentation.

Interpersonal skills

1) Lead stakeholder meetings; 2) Public-speaking skills.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

A postdoctoral scholar, two undergraduate students, four graduate students, and one part-time design development engineer were involved in conducting the experiments and other activities. The postdoc working under the guidelines of the PI and Co-PIs led the study and day to day activities from 2017 until the end of 2019. His major activities for this period included conducting a literature review, preparing the FlexiDry experimental plan, conducting experiments, prototyping the lab-scale drying system, collecting data, statistical analysis, configuring various National Instrument C-series modules and integrating them with sensors, developing LabVIEW standalone data acquisition program, purchasing equipment and instrumentation, preparing presentations and reporting. After that, a PhD student took on the postdoc's role and continued working on the project until the end of the grant period.

The design engineer works at the Biological and Agricultural Engineering (BAE) development shop, and was in continuous communication with PIs, co-PIs, the Postdoctoral scholar, and graduate students. He, alongside one undergraduate student handled any design issues that came up when using the system.

The graduate student activities for this period included reviewing literature, preparing experimental plans, sample preparation, carrying out the experiments, collecting data, analyzing data, preparing presentations, and reporting the results. The project's experiments and results were used in one M.S. thesis and is projected to be reported in two Ph.D. dissertations.

One undergraduate student worked closely with the design engineer in the manufacturing of the prototype dryer. The other undergraduate students assisted with the experiment including sample collection/handling, data recording, and presentations.



3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Information products resulting from the project are listed chronologically below.

2018

- **Poster:** Energy Efficient Desiccant-based System for Drying Agricultural Commodities. Innovator Showcase 2018, UC Davis (CA).
- **Workshop:** Implementing the Dry Chain to Reduce Postharvest Commodity Losses and Improve Food Safety. April 20, 2018. World Food Center, UC Davis (CA).

2019

- **Scientific article:** Mora A., Sufi U., Roach J. I., Thompson J., Donis-González I. R. Evaluation of a small-scale desiccant-based drying system to control corn dryness during storage. *AIMS Agriculture and Food*, 4(1):136-148, February 2019; DOI: 10.3934/agrfood.2019.1.136
- **Scientific article:** Edward, S.S., Yigal, A., Donis-González, I.R., Wendi, G., Madison, J.S., Momin, M.A., Laura M., Sara, P., Tom, Q., Kiara, W., Thomas, P.T. 2019. Food Loss and Waste: Measurement, Drivers, and Solutions (a review). *Annual Review of Environment and Resources*.
- **Poster:** Energy Efficient Desiccant-based System for Drying. Md Abdul Momin, Alice Dien, Irwin R. Donis-González, Kurt Kornbluth. 2019 Innovator Showcase, UC Davis (CA).
- **Poster:** Energy-efficient drying to reduce food loss. Johan Van Asbrouck, Irwin Donis-González, Kurt Kornbluth, Edward Spang, Pedro Bello & Kent J. Bradford. 2019 Innovator Showcase, UC Davis (CA).
- **Poster:** Energy Efficient Desiccant-based System for Drying. Md Abdul Momin, Alice Dien, Irwin R. Donis-González, Kurt Kornbluth. 2019 Picnic Day, UC Davis (CA).
- **Poster:** Evaluation of Desiccant-based Drying Systems for Agricultural Products. Alice Dien, Md Abdul Momin, Alexandra Mora, Umayr Sufi, Jedediah



I. Roach, James F. Thompson, Kurt Kornbluth, and Irwin R. Donis-González. Institute of Food Technologists (IFT) Annual Meeting and Expo (IFT19), New Orleans (LA).

- **Industry report/workshop:** Performance evaluation of the FlexiDry® for drying corn. Irwin R. Donis González, Md Abdul Momin, Alice Dien, Kurt Kornbluth. August 2019. Western Center for Agricultural Equipment (WCAE), UC Davis (CA).

2020

- **Poster:** Evaluation of the Performance of an Industrial-Scale Desiccant-Based System for Agricultural Commodities. Alice Dien, Md Abdul Momin, Laudia Anokye-Bempah, Kurt Kornbluth, and Irwin R. Donis-González. American Society for Horticultural Science Annual Meeting (ASHS 2020). Remote.

2021

- **Poster (3rd prize):** Performance evaluation of a commercially available desiccant-based seed drying system (FlexiDry®) using corn kernels (Zea Mays). Alice Dien, Md Abdul Momin, Laudia Anokye-Bempah, Kurt Kornbluth, and Irwin R. Donis-González. American Society of Agricultural and Biological (ASABE) CA-NV Section Meeting 2021. Remote.
- **Poster:** Performance evaluation of a commercially available desiccant-based seed drying system (FlexiDry®) using corn kernels (Zea Mays).). Alice Dien, Md Abdul Momin, Laudia Anokye-Bempah, Kurt Kornbluth, and Irwin R. Donis-González. American Society of Agricultural and Biological Engineers (ASABE) Annual Meeting (AIM 2021). Remote.
- **Conference presentation:** Design of a research-scale forced convection Vertical Dryer (VD) with variable air inlet conditions to evaluate agricultural commodities drying techniques. Presented by Alice Dien at American Society of Agricultural and Biological (ASABE) Annual Meeting (AIM 2021). Remote.
- **Poster:** Performance evaluation of zeolite-based desiccant beads for the convective drying of fresh almonds and walnuts. Alice Dien, Laudia Anokye-Bempah, Kurt Kornbluth, and Irwin R. Donis-González. American Society for Horticultural Science Annual Meeting (ASHS 2021). Remote.
- **Symposium Presentation:** The use of desiccants for proper moisture preservation in green coffee during storage and transportation. Laudia Anokye-Bempah, Juliet Han, William Ristenpart, Irwin R. Donis-González. UC Davis Coffee Center Research Symposium 2021.



- **Poster: The use of desiccants for proper moisture preservation in green coffee during storage and transportation.** Laudia Anokye-Bempah, Juliet Han, William Ristenpart, Irwin R. Donis-González. Specialty Coffee Expo 2021, New Orleans.

2022

- **Poster (1st prize):** Design of a research-scale forced convection vertical dryer with variable inlet conditions. Alice Dien, Laudia Anokye-Bempah, Jedediah Roach, Kurt Kornbluth and Irwin R. Donis-González. American Society of Agricultural and Biological (ASABE) CA-NV Section Meeting 2022. Remote.
- **Scientific article:** Dien A., Momin Md. A., Ying T., Spang E. S., Kornbluth K., Donis-González I. R. Performance Evaluation of a Commercially Available Desiccant-Based Seed Drying System. Journal of the ASABE. 65(3): 633-643. (doi: 10.13031/ja.14762) @2022.
- **UC Davis Grad Slam Talk (1st prize and Public Impact prize):** "Cooling down with the new hot air: The future of drying in agriculture" by Alice Dien.
- **CALESS Elevator Pitch Competition (1st prize):** The use of desiccants for proper moisture preservation in green coffee during storage and transportation by Laudia Anokye-Bempah.
- **M.S. Thesis:** Preservation and moisture estimation of green coffee by Laudia Anokye-Bempah.
- **Scientific Article (Submitted and under review):** Anokye-Bempah L., Han J., Kornbluth K., Ristenpart W., Donis-Gonzalez I. R. The use of desiccants for proper moisture preservation in green coffee during storage and transportation. Journal of Stored Products Research. 2022.

In works

- **Scientific article:** Dien A., Anokye-Bempah L., Roach J., Kornbluth K., Donis-González I.R. Design and preliminary tests of a research-scale forced-convection deep-bed dryer with variable inlet conditions.
- **Scientific article:** Ying T., Dien A., Spang E., Donis-González I. R. Drying performance and quality preservation in Calrose rice paddy (*Oryza sativa*) using convective zeolite-based desiccant drying.
- **Scientific article. Dien A.,** Anokye-Bempah L., Kornbluth K., Donis-González I. R. Drying performance and quality preservation of almonds and walnuts using convective desiccant-based drying.



- **Ph.D. dissertation (2023).** Energy and Greenhouse Gas Assessment of Current Emerging Rice Drying Technologies by Tianyu Ying.
- **Ph.D. dissertation (2024).** Feasibility Analysis of a Deep-bed Forced-convection Desiccant-based Dryer for Grain and Tree Nut Crops by Alice Dien.

3.2. Please provide a list of citations for the information products produced during the project.

1. Mora, A., Sufi, U., Roach, J. I., Thompson, J.F, Donis-González, I.R. 2018. Evaluation of a small-scale desiccant-based drying system to control corn dryness during storage. *AIMS Agriculture and Food* 4(1):136-148.
2. Edward, S.S., Yigal, A., Donis-González, I.R., Wendi, G., Madison, J.S., Momin, M.A., Laura M., Sara, P., Tom, Q., Kiara, W., Thomas, P.T. 2019. Food Loss and Waste: Measurement, Drivers, and Solutions (a review). *Annual Review of Environment and Resources*.
3. Dien A., Momin Md. A., Ying T., Spang E. S., Kornbluth K., Donis-González I. R. Performance Evaluation of a Commercially Available Desiccant-Based Seed Drying System. *Journal of the ASABE*. 65(3): 633-643. (doi: 10.13031/ja.14762) @2022.
4. Anokye-Bempah L., Han J., Kornbluth K., Ristenpart W., Donis-Gonzalez I. R. The use of desiccants for proper moisture preservation in green coffee during storage and transportation. *Journal of Stored Products Research*. 2022. [submitted and under review].

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

1. Mora, A., Sufi, U., Roach, J. I., Thompson, J.F, Donis-González, I.R. 2018. Evaluation of a small-scale desiccant-based drying system to control corn dryness during storage. *AIMS Agriculture and Food* 4(1):136-148.
2. Edward, S.S., Yigal, A., Donis-González, I.R., Wendi, G., Madison, J.S., Momin, M.A., Laura M., Sara, P., Tom, Q., Kiara, W., Thomas, P.T. 2019. Food Loss and Waste: Measurement, Drivers, and Solutions (a review). *Annual Review of Environment and Resources*.
3. Dien A., Momin Md. A., Ying T., Spang E. S., Kornbluth K., Donis-González I. R. Performance Evaluation of a Commercially Available Desiccant-Based Seed Drying System. *Journal of the ASABE*. 65(3): 633-643. (doi:

10.13031/ja.14762) @2022.

4. Anokye-Bempah L., Han J., Kornbluth K., Ristenpart W., Donis-Gonzalez I. R. The use of desiccants for proper moisture preservation in green coffee during storage and transportation. Journal of Stored Products Research. 2022. [submitted and under review].
- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.
- Lucyna Kurtyka and Renee Bullion from the FFAR visited our booth in the Innovator Summit, 2018. We shared our on-going project activities and had a good discussion about our project advancement and work. The following link and the screenshot show their expression towards us.



- The Innovation Institute for Food and Health (IIFH) at UC Davis prepared a video as part of their 2018 summer program. They considered the desiccant beads-based drying technology as an innovative technique that can transform how produce is dehydrated. It was a great opportunity to briefly talk about our project goals, and about an alternative drying technology for agricultural produce. The video is available at the following link: <https://www.youtube.com/watch?v=2BsJ6pG9Kfo#action=sha>

- Ph.D. student Alice Dien won 1st place and Public Impact Award by pitching this project at the 2022 UC Davis Grad Slam competition which rewards students for communicating research to a general audience. Video and press-release can be found in the following link: <https://grad.ucdavis.edu/news/biological-systems-engineering-grad-student-alice-dien-wins-grad-slam>
- Several blog posts and news were written after the Grad Slam competition and can be found at the following links:
 - “New Hot Air is a Grad Slam Dunk. Alice Dien Uses Little Air to Explain Her Research.” <https://www.ucdavis.edu/news/new-hot-air-grad-slam-dunk>
 - “Alice Dien Hits it Out of the Park, Advances to UC Grad Slam Finals with Revolutionary Drying Research.” <https://bae.ucdavis.edu/news/alice-dien-advances-uc-grad-slam-finals-revolutionary-drying-research>
 - “A Dry Idea to Save Energy and Preserve Food. Could Desiccation be the Future of Drying in Agriculture?” <https://www.ucdavis.edu/blog/dry-idea-save-energy-and-preserve-food>

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Conversations were initiated with the Innovation access office at the UC Davis Office of Research to apply and extend international and U.S. patents for desiccant-based drying systems. However, proof of concept and further design parameters need to be tested and demonstrated to finalize patent application.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	



- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Yes, the overall concept and design of our desiccant system, including schematic representations. Please, describe the generalities of our study, but do not offer specifics about our approach and design schematics. These will be shared through scientific publications.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

We have created a dry-chain workgroup in collaboration with the World Food Center, the Agricultural and Natural Resources, and the Postharvest technology center at UC Davis. In addition, we have a working agreement with the Innovation Institute for Food and Health (IIFH) at UC Davis. All these institutions are committed to catalyzing the innovation across food, agriculture and health in the United States and around the world.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

Yes.

1. We evaluated the performance of industry scale desiccant beads-based seed drying system using heated and unheated air drying. We compared these with conventional heated-air drying. Performance evaluation included drying capacity, rate of moisture removal, drying efficiency, energy usage, and quality evaluation.
2. Drying process characterization of walnuts, almonds, and rice using heated air drying and Drying Beads. Data includes temperature and RH changes at different points within the process. Weight changes to plot drying curves at different process conditions. Airflow.
3. We investigated the efficacy of commercially available desiccants for preserving the moisture content of green coffee between its optimal range (10-12%) when stored in either hermetic packages and/or jute sacks and assessed the corresponding impact on sensory quality.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

As described above, the generation of data through a specific milestone defined future data sets and experimental design. So, overall all of the data generated in the project is related.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

No relevant metadata has been generated, but only transitional data that needs to be analyzed and summarized accordingly before providing copies. Metadata will be safely kept internally in safe servers (Box folders), and will be forwarded to FFAR as soon as it is available.



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Irwin R. Donis-González

PI Signature: Irwin R Donis Gonzalez Date: August 23, 2022

Authorized Signing Official (ASO) Name: Grace I. Liu, J.D.
Associate Director, Sponsored Programs

ASO Signature: Grace I Date: January 23, 2023